

NICOLAS CEZAR DE CARVALHO BERREDO

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EDUCATION & SKILLS

B.S. Electrical and Computer Engineering | University of Washington, Seattle, WA

Expected Graduation: June 2028 GPA: 3.52

Relevant Coursework: Signals & Systems, Electromagnetism, Data Structures & Algorithms, Intermediate Data Programming

In Progress: Digital Circuits, Exploring Hardware Devices

Dean’s List: Fall 2024, Spring 2025, Fall 2025

Languages: Portuguese (native), English (fluent), Spanish (intermediate).

Skills: Embedded systems, prototyping, testing; Python, Java; SPI, I2C, CAN J1939; Arduino; KiCad, SolidWorks, LTspice; Oscilloscope, Soldering; Linux

ENGINEERING EXPERIENCE

R&D Electrical Team Member | E-Truck Challenge @ UW | Seattle, WA 2026-Present

- Designed CAN J1939 message definitions in CANdb++ for a radiator-mounted cooling fan.
- Prototyped control and safety-interlock circuits for Project RAIJIN’s high-voltage system using comparators, regulators, transducers, and timers; performed bench testing with oscilloscopes and power supplies to verify interlock voltage.

Undergraduate Research Assistant | UW ARC Lab - Project SHIELD | Seattle, WA 2026-Present

- Built an end-to-end Python pipeline for sensor health monitoring, generating 150+ GB of synthetic fault data through sensor-specific Gaussian noise injection across multiple types of sensors.
- Implemented signal-processing and machine-learning pipelines using MODWT, time-frequency features, Random Forest, and XGBoost to model sensor degradation and assess hardware faults.
- Parallelized sensor-data processing, achieving 3.1x faster execution for large-scale analysis of recordings.

Electronics Subteam Member | Reptiles PUC-Rio Baja SAE Team | Rio de Janeiro, Brazil 2024

- Contributed to the electronics of a BAJA SAE off-road motorsports team, helping win a regional championship
- Prototyped an embedded vibration-measurement system integrating an accelerometer, MCU, and SD module via I2C and SPI to evaluate suspension configurations and assess driver comfort following ISO 2631-1
- Used RapidHarness to visualize wiring harness and SolidWorks CAD tools to design a vehicle stand for maintenance and inspection

Project Manager | Smart Recycling Conveyor Belt Project, PUC-Rio | Rio de Janeiro, Brazil 2024

- Built a smart recycling conveyor proof-of-concept as part of a waste-management study focused on the Parque da Cidade community in Rio de Janeiro.
- Prototyped and programmed an Arduino-based embedded control system integrating a proximity sensor, relay, LEDs, activation button, and external power supply.
- Managed the project using Scrum and earned the “Most Outstanding Student” distinction.

WORK EXPERIENCE

Operations & Technology Summer Intern | BTG Pactual Asset Management | Rio de Janeiro, Brazil 2026

- Developed a Python optimization and simulation framework for target-date retirement ETFs, modeling dynamic equity and fixed-income allocations.
- Built an interactive dashboard for portfolio analytics and scenario visualization to support new financial products.

AI Developer Summer Intern | Maximiza Tecnologias | Rio de Janeiro, Brazil 2025

- Led development of a WhatsApp ordering chatbot powered by OpenAI’s LLM, supporting more than 100 customer inquiries daily.
- Developed a Streamlit application with an AI assistant to manage orders, query data, and automate notifications.

LEADERSHIP ACTIVITIES

President | Brazilian Student Association at UW | Seattle, WA 2026-Present

Co-Founder & Director of Logistics 2025-2026

- Built Notion dashboards to coordinate event planning and team handoffs; formalized the association’s constitution and bylaws.
- Planned 8+ events with 50 attendees each and raised US\$1,500 while supporting 50+ Brazilian students in Seattle.